Lab 9: Creating Different Network Topologies using Packet Tracer

Assignment Date: 19 July 2026 | Deadline: 27 July 2026

BSc.CSIT 4th Semester | New Summit College | Computer Network

Objective: To understand the concept of network topology and implement Ring, Star, and Mesh topologies in Cisco Packet Tracer.

1. Theory

1.1 Different Topologies

A network topology refers to the arrangement of devices and connections in a network. It can be physical (actual cable layout) or logical (how data actually flows). The topology chosen affects a network's cost, reliability and ease of troubleshooting.

- **Ring Topology:** Each device is connected to exactly two other devices, forming a circular path for data. Data travels around the ring until it reaches its destination; failure of one link/device can disrupt the whole ring unless a dual-ring or redundancy mechanism is used.
- **Star Topology:** All devices are connected individually to a central device (switch/hub). It is easy to install and manage, and failure of one device does not affect the others, but the central device becomes a single point of failure.
- **Mesh Topology:** Every device is connected to every other device, providing multiple paths for data and high redundancy/fault tolerance, at the cost of requiring many more cables and ports as the network grows.

1.2 Components Used

- End Devices (PCs) — as the nodes of each topology.
- Switches/Hubs — used as connecting devices for the star topology.
- Copper Straight-Through and Crossover Cables — used to interconnect devices depending on topology.

1.3 Network Diagrams

■ ***[Insert Screenshot Here]** — network diagram of the Ring topology*

■ ***[Insert Screenshot Here]** — network diagram of the Star topology*

■ ***[Insert Screenshot Here]** — network diagram of the Mesh topology*

2. Implementation Sequence

(a) Ring Topology

a. Place the required PCs in the workspace and connect each PC to exactly two neighboring PCs using crossover cables, forming a closed loop.

b. Assign static IP addresses to each PC on the same subnet.

c. Test connectivity between two devices using the `ping` command.

■ ***[Insert Screenshot Here]** — completed Ring topology with successful ping between two devices*

(b) Star Topology

a. Place a central switch and connect each PC individually to the switch using straight-through cables.
b. Assign static IP addresses to each PC on the same subnet.
c. Test connectivity between two devices using the `ping` command.

■ ***[Insert Screenshot Here]*** *— completed Star topology with successful ping between two devices*

(c) Mesh Topology

a. Place the required PCs and connect every PC to every other PC using crossover cables.
b. Assign static IP addresses to each PC on the same subnet.
c. Test connectivity between two devices using the `ping` command.

■ ***[Insert Screenshot Here]*** *— completed Mesh topology with successful ping between two devices*

Conclusion

Implementing the Ring, Star and Mesh topologies in Packet Tracer helped in understanding how the physical arrangement of devices affects cabling requirements, fault tolerance and manageability of a network. Testing connectivity in each topology confirmed that all devices could communicate correctly once properly connected and addressed.

Lab 10: Creating VLAN and VLAN Trunking using Packet Tracer

Assignment Date: 19 July 2026 | Deadline: 27 July 2026

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Objective: To understand the concept of VLAN and VLAN Trunking, and implement them in Cisco Packet Tracer.

1. Theory

1.1 VLAN, VLAN Trunking & its Architecture

A VLAN (Virtual Local Area Network) is a logical grouping of devices on one or more switches that behaves as if it were on its own separate physical LAN, even though the devices may be connected to the same physical switch. VLANs are used to segment traffic for security, performance and easier management, without needing separate physical switches for each group.

VLAN Trunking allows traffic from multiple VLANs to travel over a single physical link between two switches. A trunk port tags each frame with a VLAN ID (commonly using the IEEE 802.1Q standard) so that the receiving switch can identify which VLAN the frame belongs to and forward it accordingly.

1.2 Components Used

- Two Switches — configured with VLANs and a trunk link between them.
- End Devices (PCs) — assigned to different VLANs on each switch.
- Straight-Through Cable (PC to Switch) and Crossover Cable (Switch to Switch trunk link).

1.3 Network Diagram

■ ***[Insert Screenshot Here]*** *— network diagram showing two switches, their VLANs, PCs and the trunk link*

2. Implementation Sequence

2.1 Configuring VLANs and VLAN Trunking

a **Create VLAN on Switch 1 & assign port:** On Switch 1's CLI, use `vlan <id>` and `name <vlan-name>` to create the VLAN, then assign the relevant access port(s) to it using `switchport mode access` and `switchport access vlan <id>`.

b **Create VLAN on Switch 2 & assign port:** Repeat the same VLAN creation and port assignment steps on Switch 2 for its connected PCs.

c **Configure trunking between switches:** On the port connecting the two switches, set `switchport mode trunk` on both ends so that tagged traffic from all VLANs can pass between the switches.

■ ***[Insert Screenshot Here]*** *— VLAN creation and port assignment commands on Switch 1*

■ ***[Insert Screenshot Here]*** *— VLAN creation and port assignment commands on Switch 2*

■ ***[Insert Screenshot Here]*** *— trunk port configuration between the two switches*

2.2 Testing and Validation

After configuration, connectivity is verified by pinging between PCs in the same VLAN (should succeed) and between PCs in different VLANs (should fail, confirming traffic isolation), as well as by using `show vlan brief` on each switch to confirm correct port-to-VLAN assignment.

■ ***[Insert Screenshot Here]*** *— successful ping between two PCs in the same VLAN*

■ ***[Insert Screenshot Here]*** *— show vlan brief output confirming VLAN assignments*

2.3 Addressing Table

Device	Interface	IP Address	Subnet Mask	VLAN
PC1	NIC	192.168.10.1	255.255.255.0	10
PC2	NIC	192.168.10.2	255.255.255.0	10
PC3	NIC	192.168.20.1	255.255.255.0	20
PC4	NIC	192.168.20.2	255.255.255.0	20

Note: Replace the sample values above with the actual IPs/VLANs used in your own topology (refer to KEC Book, page 264, for the sample format).

Conclusion

This lab provided practical understanding of how VLANs logically segment a switched network and how VLAN trunking allows multiple VLANs to share a single inter-switch link. Testing showed that devices in the same VLAN could communicate freely while devices in different VLANs remained isolated, confirming correct VLAN and trunk configuration.